INCH-POUND

MIL-C-3098/53J
10 April 1995
SUPERSEDING
MIL-C-3098/53H
9 November 1983

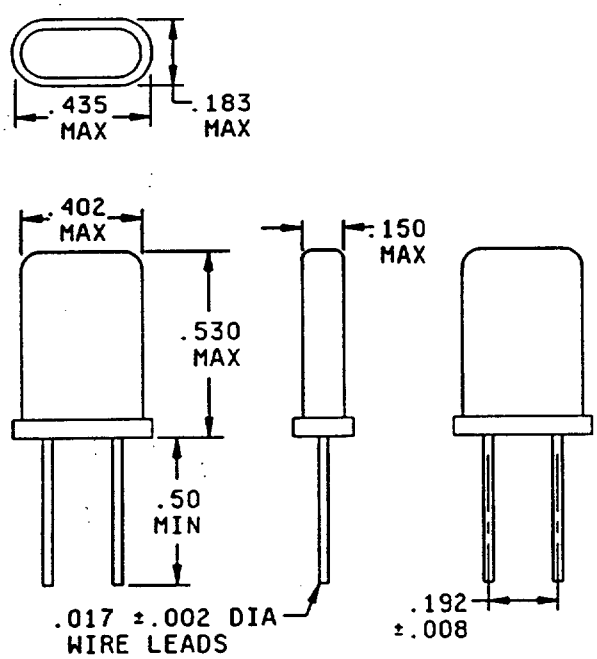
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-76/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation:
MIL-C-3098.

Pertinent characteristics: 16 MHz to 61 MHz; third overtone; noncontrolled; series.



Inches	mm
.002	0.05
.008	0.20
.017	0.43
.150	3.81
.183	4.65
.192	4.88
.402	10.21
.435	11.05
.50	12.7
.530	13.46

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking to be in accordance with MIL-C-3098.
4. If the HC-43 holder is used the crystal unit shall not be trimmed after cold-welding.

FIGURE 1. Crystal unit - CR-76/U.

AMSC N/A

FSC 5955

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:33Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Holder: HC-18/U or HC43/U (MIL-H-10056/9) or HC-49/U (MIL-H-10056/36).

Dimensions, marking, and configuration: See figure 1.

Frequency range: 16 MHz to 61 MHz, inclusive.

Capacitance, shunt: 3.5 pF, maximum.

Frequency tolerances:

Primary (operating temperature range): ±20 ppm.

Secondary (operating temperature range): ±30 ppm.

Equivalent resistance: 40 ohms, maximum.

Mode of oscillation: Third mechanical overtone.

Temperature ranges:

Primary: -40°C to +90°C, inclusive.

Secondary: -55°C to -40°C and +90°C to +105°C, inclusive.

Drive level, rated: 2.0 ±0.4 mW.

Unwanted modes: Twice main mode.

Shock (specified pulse):

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Vibration high frequency: MIL-STD-202, method 204, test condition A.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Thermal shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Aging:

Frequency change permitted: ±5 ppm.

NOTE: NATO preferred type NXT51.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:33Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC, OS, SH
Air Force - 17, 19, 99
DLA - ES

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5955-0680-03)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:33Z
Check the source to verify that this is the current version before use.